

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 20

CLASSIFICATION OF WINDS

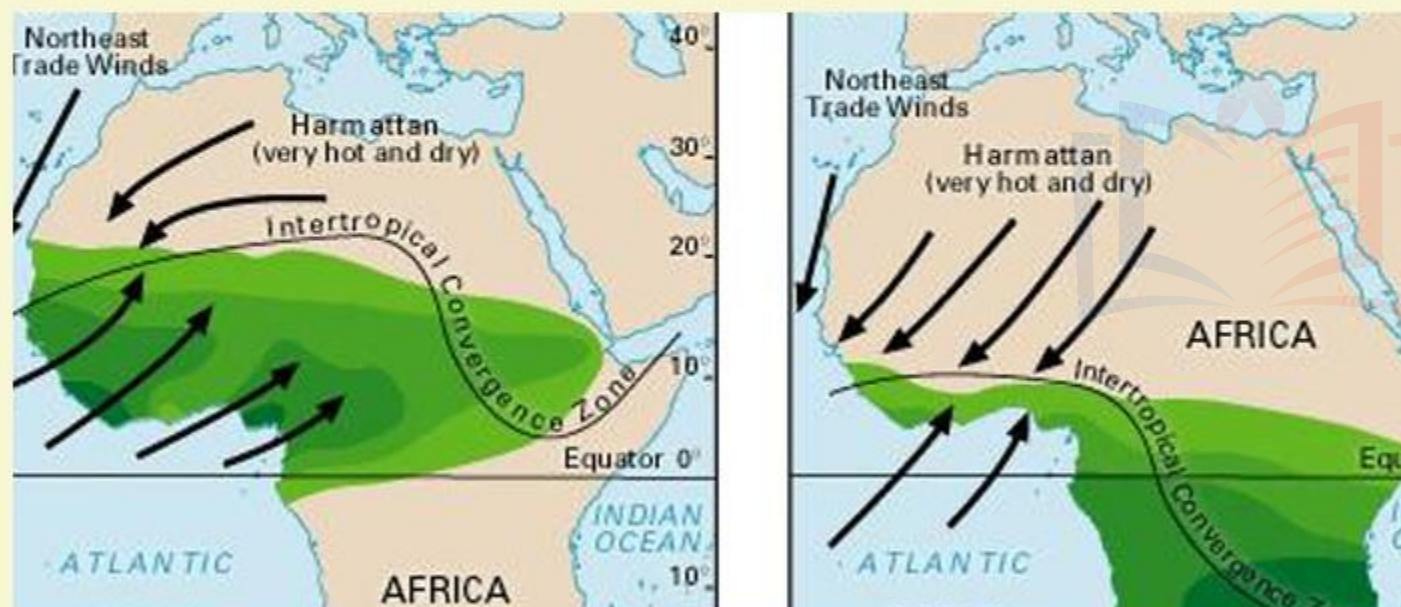
- **LOCAL WINDS(CONTD..)**

CLASSIFICATION OF WINDS

LOCAL WINDS

4

HARMATTAN



1 The warm and dry winds blowing from north-east and east to west in the parts of Sahara Desert are called Harmattan.

2 These winds become extremely dry because of their journey over Sahara desert.

CLASSIFICATION OF WINDS

LOCAL WINDS

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HARMATTAN



3 The western coast of Africa is warm and moist and hence the weather becomes unpleasant because the weather conditions characterised by high relative humidity become injurious for human health

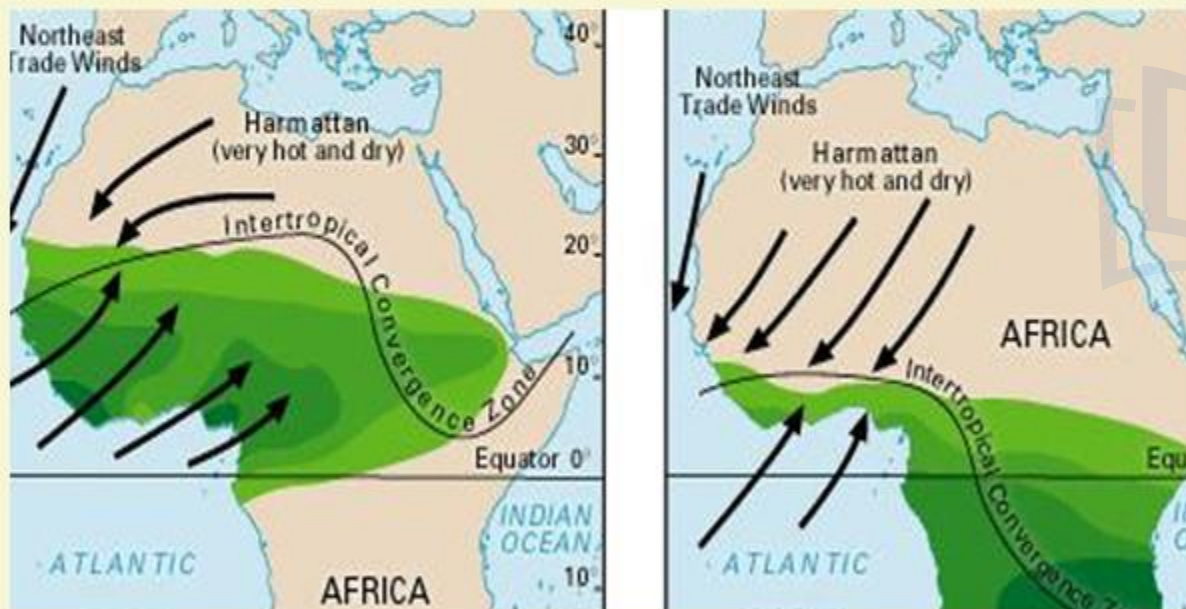
4 With the arrival of Harmattan the weather becomes dry and pleasant as the relative humidity of the air remarkably reduced due to high temperature and hyperaridity of Harmattan.

CLASSIFICATION OF WINDS

LOCAL WINDS

4

HARMATTAN



5 This is why Harmattan is known as “**Doctor**” in the Guinea coastal area of western Africa

6 In fact, harmattan winds are very dusty and stormy winds blowing with high speed that trees are uprooted.

7 These winds are usually associated with **dust storms** resulting into **marked reduction in the visibility**. The winds become more vigorous during summer months.

8 Harmattan is **special type of north-east trade wind**.

CLASSIFICATION OF WINDS

LOCAL WINDS

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HARMATTAN

OTHER NAMES OF HARMATTAN

Similar warm, dry, very strong and dust laden winds are called by different names in different places.

NAME OF THE WINDS	PLACES
Brickfielder	Victoria Province of Australia
Blackroller	Great Plains of the USA
Norwester	New Zealand
Shamal	Mesopotamia and Persian Gulf

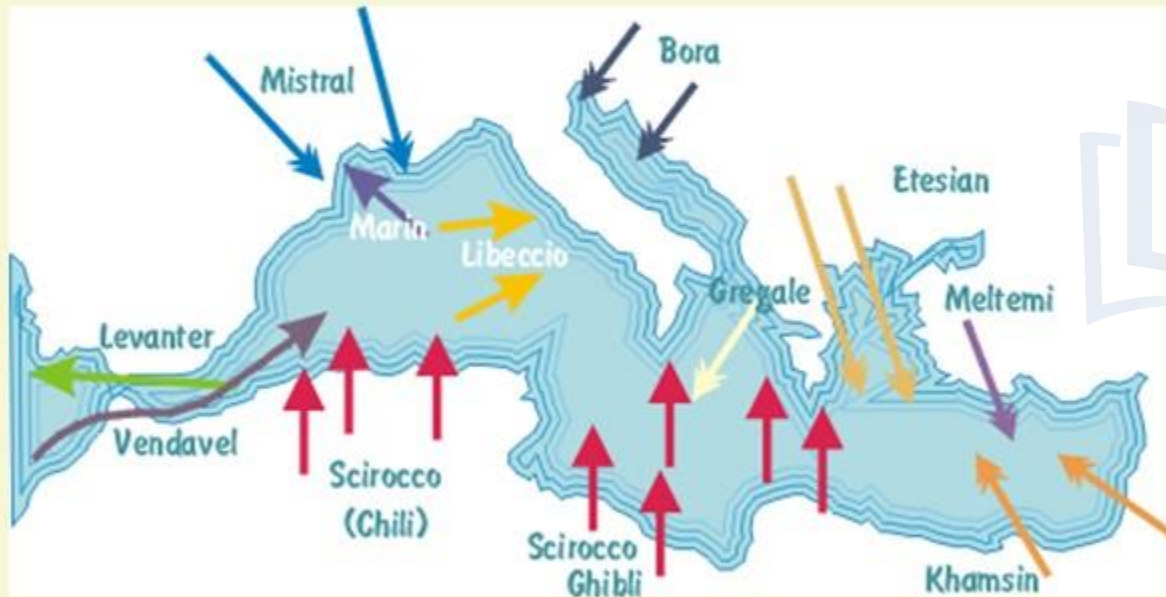


CLASSIFICATION OF WINDS

LOCAL WINDS

5

SIROCCO



1 Sirocco is warm, dry and dusty local winds, which blows in northerly direction from Sahara desert and after crossing over the Mediterranean sea it reaches Italy, Spain etc.

2 Sirocco becomes very strong and active at the time of origin of cyclonic storms over the Mediterranean Sea.

3 Sirocco, while passing over the Mediterranean Sea picks up moisture and yields rainfall in the southern part of Italy, where the rain associated with Sirocco is called “Blood Rain” because of fallout of red sands with falling rains.

CLASSIFICATION OF WINDS

LOCAL WINDS

5

SIROCCO

OTHER NAMES OF SIROCCO

NAME OF THE WINDS	PLACES
Khamsin	Egypt
Chilli	Tunisia
Gibli	Lybia

4

It may be remembered that Sirocco, while blowing through Sahara desert it **picks up red sands** which settles down with rains in South Italy.

5

It is apparent that Sirocco is very much **injurious to agricultural and fruits crops**.

CLASSIFICATION OF WINDS

LOCAL WINDS

6

MISTRAL



1 Mistral is a **cold local wind** which blows in **Spain and France** from north-west to south-east direction.

2 These winds are more common and **effective during winter season** because of development of high pressure over Europe and low pressure over Mediterranean Sea.

3 These winds become extremely cold when they blow through central plateau and **descend into Rhone valley** on the southern coast of France.

CLASSIFICATION OF WINDS

LOCAL WINDS

6

MISTRAL



4 The average velocity is around 60 km per hour, but sometimes it can reach even up to 130 km per hour.

5 These stormy winds adversely affect airflights.

6 The arrival of mistral causes sudden drop in air temperature to below freezing point.

Mistrals are a cold, northerly wind flowing into the Gulf of Lion from the southern coast of France, are most common during winter and spring.

CLASSIFICATION OF WINDS

LOCAL WINDS

7

BLIZZARD

- 1 Blizzard is very violent, stormy, cold and powdery polar wind laden with dry snow and it is prevalent in polar regions of Siberia, Canada and USA.
- 2 The visibility becomes remarkably low because of snow and ice crystals.
- 3 The arrival of these winds causes sudden drop in air temperature to subfreezing level, thick cover of snow on the ground surface and onset of cold waves.
- 4 These winds reach the southern States of USA because of the absence of any east-west mountain barriers in the USA.
- 5 They are also called as “norther” in southern USA and “Burran” in Siberia.



CLASSIFICATION OF WINDS

LOCAL WINDS



OTHER LOCAL WINDS

- 1 **Levanter**- A strong easterly cold wind in **Southern Spain**
- 2 **Norwester**- A warm, dry and gusty wind in **New Zealand**
- 3 **Pampero**- A north-westerly cold wind in the “**Pampas**” of South America
- 4 **Santa Ana**- A warm and dry wind in the **USA**
- 5 **Zonda**- A warm wind in **Argentina**

CLASSIFICATION OF WINDS

LOCAL WINDS

PROBABLE QUESTION FOR MAINS

Q 3 WHAT ARE LOCAL WINDS? HOW DO THEY AFFECT THE LOCAL WEATHER CONDITIONS AND HUMAN HABITATS? **(10 Marks) (150 words)**

HINTS:

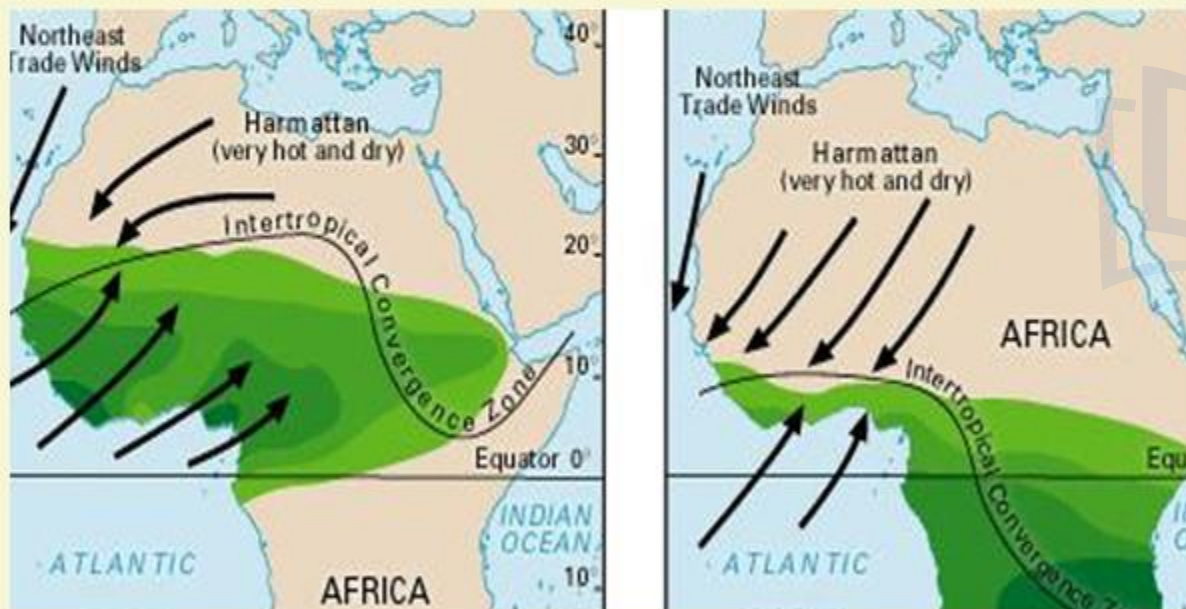
1. Straight forward question. Go with the theory part we have dealt.
2. Define local winds.
3. Thermal factors and topographical factors.
4. Write second half answer with the NAMES AND PLACES of local winds.
5. Chinook, Foehn- Snow eater; 4 degree Celsius rise; wheat sowing; grapes ripening.
6. Sirocco- Blood Rain; Injurious to agri crops.
7. Blizzard- Cold snow storm even upto southern states of US.
8. Similarly, Harmattan also can be quoted as an example.

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